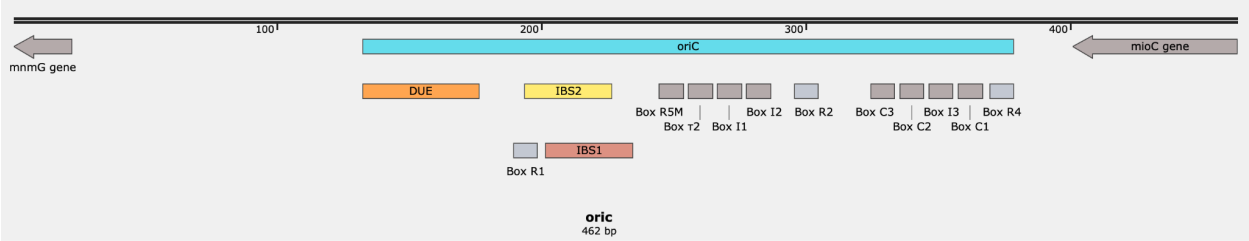
oriC

The full oriC sequence is shown here: [oriC sequence]. In *E. coli*, oriC consists of two functional regions: the duplex unwinding element (DUE) and the DnaA-oligomerization region (DOR). The DUE is the site at which the DNA double helix is locally unwound to permit replisome loading, whereas the DOR serves as the platform for DnaA binding and oligomerization that drives DUE opening.



Within the DOR, there are 11 DnaA boxes. Three of these—R1, R2, and R4—are high-affinity sites that bind both DnaA–ATP and DnaA–ADP, with dissociation constants on the order of 1 nM. Among them, R1 and R4 are the highest-affinity consensus boxes (TTATCCACA), whereas R2 (TTATACACA) has slightly lower affinity. The remaining eight sites (R5M, τ2, I1, I2, C3, C2, I3, and C1) are low-affinity sites that preferentially require DnaA–ATP and cooperative interactions for stable occupancy; their individual binding affinities are weaker than 100 nM. In addition to DnaA boxes, the DOR contains binding sites for the nucleoid-associated protein IHF (integration host factor). The primary IHF-binding site, IBS1, is located between R1 and R5M, whereas the secondary site, IBS2, overlaps with R1. It has been suggested that IHF binds mainly to IBS1 when R1 is occupied by DnaA.

Initiation at oriC is thought to proceed as follows. Before initiation, the three high-affinity boxes are occupied by DnaA bound to either ATP or ADP, while IHF binds to IBS1. Initiation begins in the right half of the DOR, where DnaA–ATP cooperatively assembles in an ordered manner on the low-affinity sites C1, I3, C2, and C3, in association with R4-bound DnaA. IHF-induced bending of the DNA then promotes formation of a second DnaA–ATP filament associated with R1-bound DnaA on the left half of the DOR. R2-bound DnaA is also thought to facilitate cooperative assembly. Together, these DnaA oligomers generate the torsional strain that promotes local unwinding of the adjacent DUE. The opened DUE then provides the single-stranded DNA substrate required for DnaB helicase loading and subsequent replisome assembly.

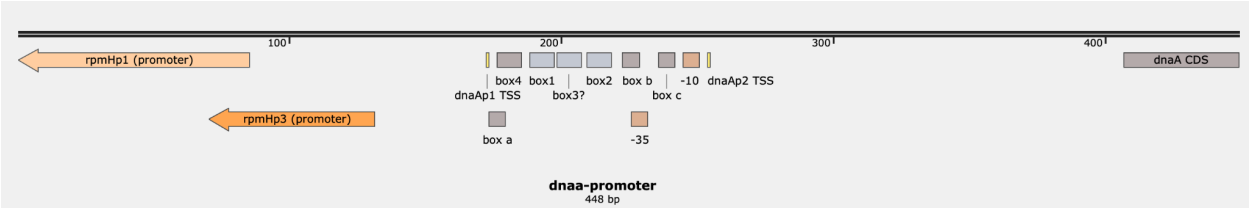
Immediately after initiation, the DnaA complex is disassembled. To prevent rapid rebinding of DnaA to newly replicated oriC, the SeqA sequestration mechanism plays an important role. In *E. coli*, Dam methyltransferase methylates the adenine residue within GATC sequences on both DNA strands. Following initiation and semiconservative replication, oriC becomes transiently hemimethylated. During this period, which lasts about 10 minutes (roughly one-third of a

doubling time under rapid growth), SeqA binds to more than ten GATC sites in newly replicated oriC and forms multimers, thereby sequestering oriC and preventing DnaA rebinding and reinitiation.

Key references:
[1] Katayama, Tsutomu, Kazutoshi Kasho, and Hironori Kawakami. "The DnaA cycle in Escherichia coli: activation, function and inactivation of the initiator protein." *Frontiers in Microbiology* 8 (2017): 2496.
[2] Kasho, Kazutoshi, Shogo Ozaki, and Tsutomu Katayama. "IHF and Fis as Escherichia coli cell cycle regulators: activation of the replication origin oriC and the regulatory cycle of the DnaA initiator." *International journal of molecular sciences* 24.14 (2023): 11572.

dnaA promoter

The full **dnaA promoter** sequence is shown here: *dnaA promoter sequence*, which is the sequence used in previous reconstruction studies. The *dnaA* gene has two promoters, p1 and p2, separated by approximately 80 base pairs. The p1 promoter provides a basal level of transcription, whereas p2 is about three-fold stronger than p1.



Between p1 and p2 lie several DnaA-binding sites, including boxes 1–4 and boxes a–c. Box 1 (TTATCCACA) is a consensus DnaA box with the highest binding affinity for both DnaA–ATP and DnaA–ADP. Box 2 differs from the consensus sequence by one base pair and is also considered a high-affinity binding site for both nucleotide-bound forms of DnaA. Box 3 has been suggested to be a very low-affinity site for DnaA–ATP/ADP. Box 4 overlaps with box a and appears to be bound predominantly by DnaA–ATP. Boxes b and c are also DnaA–ATP-preferential sites, with box b having the higher affinity of the two.

Binding of DnaA to these sites gives rise to complex autoregulatory behavior. Although most studies have concluded that DnaA represses transcription from its own promoter, a more recent study reported that promoter p2 can be subject to both positive and negative regulation by DnaA.

The *dnaA* promoter region is also rich in GATC sites. After replication of this locus, SeqA binds transiently to hemimethylated GATC sites, thereby modulating *dnaA* expression for a short period. This regulation may help buffer the transient increase in *dnaA* gene dosage caused by replication.

Key references:
[1] Speck, Christian, Christoph Weigel, and Walter Messer. "ATP–and ADP–DnaA protein, a molecular switch in gene regulation." *The EMBO journal* 18.21 (1999): 6169-6176.

[2] Saggioro, Chiara, Anne Olliver, and Bianca Sclavi. "Temperature-dependence of the DnaA–DNA interaction and its effect on the autoregulation of dnaA expression." *Biochemical Journal* 449.2 (2013): 333-341.

DnaA boxes

The consensus DnaA box sequence is **TTWTNCACA**, where **W** represents **A** or **T**, and **N** can be any nucleotide. Among these variants, **TTATCCACA** has the highest affinity for DnaA, with a dissociation constant of approximately 1 nM. More specifically, **A** is favored over **T** at the third position, and at the fifth position the preference is **C > A ≥ G > T**. DnaA boxes also exhibit cooperative behavior when two or more sites are positioned close to one another. As a result, the effective affinity of a given 9-bp sequence can also depend on its surrounding context. For example, the *mioC* promoter region contains approximately 5–8 DnaA boxes, only one of which matches the consensus sequence. In a recent bioinformatic study based on DnaA boxes found in *datA*, DARS1, and DARS2 (see below), a more relaxed motif, **HHMTHCWVH**, was adopted, where **H** represents **A, C, or T**, **M** represents **A or C**, **W** represents **A or T**, and **V** represents **A, C, or G**.

Key references:
[1] Schaper, Sigrid, and Walter Messer. "Interaction of the initiator protein DnaA of Escherichia coli with its DNA target." *Journal of Biological Chemistry* 270.29 (1995): 17622-17626.
[2] Roth, Angelika, and Walter Messer. "High-affinity binding sites for the initiator protein DnaA on the chromosome of Escherichia coli." Molecular microbiology 28.2 (1998): 395-401.
[3] Hansen, Flemming G., et al. "Insights into the quality of DnaA boxes and their cooperativity." *Journal of molecular biology* 355.1 (2006): 85-95.
[4] Olivi, Lorenzo, et al. "The Escherichia coli replication initiator DnaA is titrated on the chromosome." *Nature Communications* 16.1 (2025): 7813.

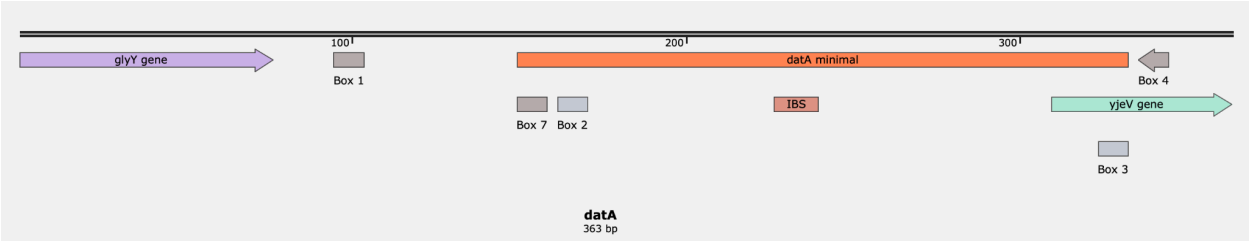
RIDA

RIDA (regulatory inactivation of DnaA) is a replication-coupled mechanism triggered by the DNA-loaded β clamp of the DNA polymerase III holoenzyme. After initiation, replisomes assemble and the β clamp is loaded onto newly synthesized DNA on both leading and lagging strands to support processive replication; during lagging-strand synthesis, clamps can remain on nascent DNA after Okazaki fragment completion. This DNA-bound clamp interacts with ADP-bound Hda, an AAA+ protein whose N-terminus contains a clamp-binding motif, forming a clamp–Hda complex that catalytically stimulates hydrolysis of DnaA-ATP to DnaA-ADP. In this way, RIDA couples ongoing DNA replication to the inactivation of DnaA, helping prevent reinitiation within the same cell cycle.

Key references:
[1] Katayama, Tsutomu, Kazutoshi Kasho, and Hironori Kawakami. "The DnaA cycle in Escherichia coli: activation, function and inactivation of the initiator protein." *Frontiers in Microbiology* 8 (2017): 2496.
[2] Riber, Leise, et al. "Multiple DNA binding proteins contribute to timing of chromosome replication in E. coli." *Frontiers in molecular biosciences* 3 (2016): 29.

DDAH

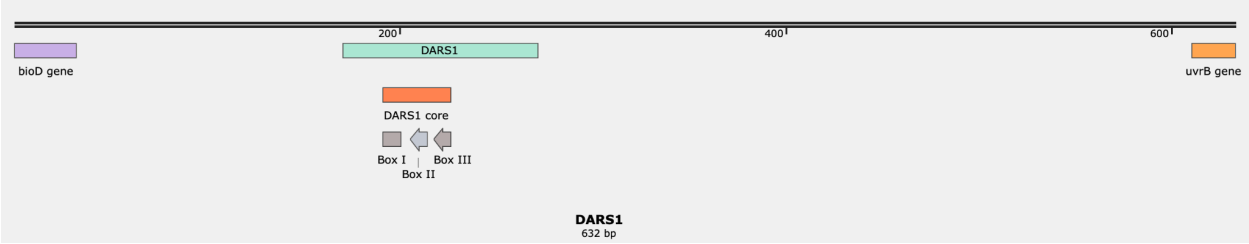
DDAH (*datA*-dependent DnaA-ATP hydrolysis) is a backup pathway to RIDA that also promotes efficient inactivation of DnaA-ATP. It depends on the *datA* locus, whose sequence is shown here: *datA sequence*. *datA* is located near oriC at 94.7 min on the *E. coli* chromosome, where binding of IHF induces a DNA architecture conducive to DnaA-ATP hydrolysis, likely through loop formation. The minimal functional *datA* region contains a single IBS and four DnaA boxes, among which boxes 2, 3, and 7 are essential and box 4 has a stimulatory role; proper spacing between the IBS and the essential DnaA boxes is also critical for activity. The resulting IHF–*datA* complex interacts catalytically with DnaA-ATP bound at the site and promotes hydrolysis of DnaA-bound ATP, and deletion of *datA* causes untimely initiation in growing cells.

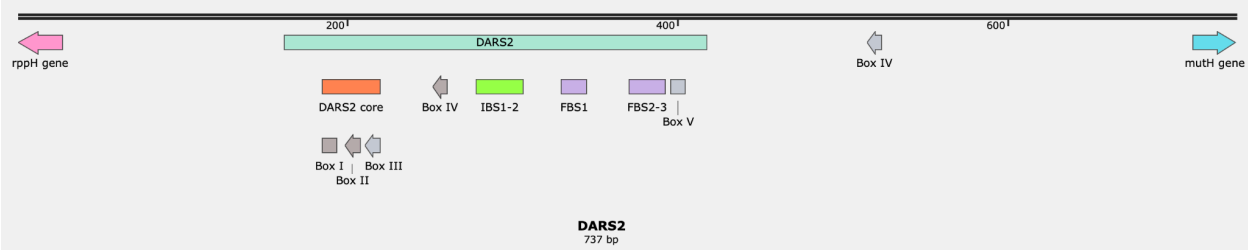


Key references:
[1] Katayama, Tsutomu, Kazutoshi Kasho, and Hironori Kawakami. "The DnaA cycle in Escherichia coli: activation, function and inactivation of the initiator protein." *Frontiers in Microbiology* 8 (2017): 2496.
[2] Hansen, Flemming G., and Tove Atlung. "The DnaA tale." *Frontiers in microbiology* 9 (2018): 319.

DARS1 & DARS2

DARS (DnaA-reactivating sequences) is the regulatory system that converts inactive ADP-DnaA back into active ATP-DnaA. It depends on two specific non-coding chromosomal loci, DARS1 and DARS2, whose sequences are shown here: *DARS1 sequence*, *DARS2 sequence*. Each of the two loci contains a characteristic set of DnaA boxes—boxes I, II, and III—that form a common core region for DnaA interaction. Within the DARS1/DARS2 complexes, ADP dissociates from DnaA-ADP, generating apo-DnaA, which can then bind ATP and thus regain initiator activity. Among the two loci, DARS2 plays the dominant role *in vivo* and contains additional binding sites for IHF and Fis; its activity is regulated during the cell cycle through the timely binding of these factors to IBS1-2 and FBS2-3, respectively, thereby promoting coordinated reactivation of DnaA.





Key references:
[1] Fujimitsu, Kazuyuki, Takayuki Senriuchi, and Tsutomu Katayama. "Specific genomic sequences of E. coli promote replicational initiation by directly reactivating ADP-DnaA." *Genes & development* 23.10 (2009): 1221-1233.
[2] Kasho, Kazutoshi, et al. "Timely binding of IHF and Fis to DARS2 regulates ATP–DnaA production and replication initiation." *Nucleic acids research* 42.21 (2014): 13134-13149.